$|a_n(z)| \leq c_n$. Then, we observe that the estimates are actually uniform in z because the c_n's are constants. It follows that the product converges uniformly to a holomorphic function, which we denote by $F(z)$.

To establish the second part of the theorem, suppose that K is a compact subset of Ω, and let

$$G_N(z) = \prod_{n=1}^{N} F_n(z).$$

We have just proved that $G_N \to F$ uniformly in Ω, so by Theorem 5.3 in Chapter 2, the sequence $\{G'_N\}$ converges uniformly to F' in K. Since G_N is uniformly bounded from below on K, we conclude that $G'_N/G_N \to F'/F$ uniformly on K, and because K is an arbitrary compact subset of Ω, the limit holds for every point of Ω. Moreover, as we saw in Section 4 of Chapter 3

$$\frac{G'_N}{G_N} = \sum_{n=1}^{N} \frac{F'_n}{F_n},$$

so part (ii) of the proposition is also proved.

3.2 Example: the product formula for the sine function

Before proceeding with the general theory of Weierstrass products, we consider the key example of the product formula for the sine function:

$$\frac{\sin \pi z}{\pi} = z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right). \tag{3}$$

This identity will in turn be derived from the sum formula for the cotangent function ($\cot \pi z = \cos \pi z / \sin \pi z$):

$$\pi \cot \pi z = \sum_{n=-\infty}^{\infty} \frac{1}{z+n} = \lim_{N\to\infty} \sum_{|n|\leq N} \frac{1}{z+n} = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2 - n^2}. \tag{4}$$

The first formula holds for all complex numbers z, and the second whenever z is not an integer. The sum $\sum_{n=-\infty}^{\infty} 1/(z+n)$ needs to be properly understood, because the separate halves corresponding to positive and negative values of n do not converge. Only when interpreted symmetrically, as $\lim_{N\to\infty} \sum_{|n|\leq N} 1/(z+n)$, does the cancellation of terms lead to a convergent series as in (4) above.

We prove (4) by showing that both $\pi \cot \pi z$ and the series have the same structural properties. In fact, observe that if $F(z) = \pi \cot \pi z$, then F has the following three properties:

(i) $F(z+1) = F(z)$ whenever z is not an integer.

(ii) $F(z) = \dfrac{1}{z} + F_0(z)$, where F_0 is analytic near 0.

(iii) $F(z)$ has simple poles at the integers, and no other singularities.

Then, we note that the function

$$\sum_{n=-\infty}^{\infty} \frac{1}{z+n} = \lim_{N\to\infty} \sum_{|n|\leq N} \frac{1}{z+n}$$

also satisfies these same three properties. In fact, property (i) is simply the observation that the passage from z to $z+1$ merely shifts the terms in the infinite sum. To be precise,

$$\sum_{|n|\leq N} \frac{1}{z+1+n} = \frac{1}{z+1+N} - \frac{1}{z-N} + \sum_{|n|\leq N} \frac{1}{z+n}.$$

Letting N tend to infinity proves the assertion. Properties (ii) and (iii) are evident from the representation $\frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2-n^2}$ of the sum.

Therefore, the function defined by

$$\Delta(z) = F(z) - \sum_{n=-\infty}^{\infty} \frac{1}{z+n}$$

is periodic in the sense that $\Delta(z+1) = \Delta(z)$, and by (ii) the singularity of Δ at the origin is removable, and hence by periodicity the singularities at all the integers are also removable; this implies that Δ is entire.

To prove our formula, it will suffice to show that the function Δ is bounded in the complex plane. By the periodicity above, it is enough to do so in the strip $|\text{Re}(z)| \leq 1/2$. This is because every $z' \in \mathbb{C}$ is of the form $z' = z + k$, where z is in the strip and k is an integer. Since Δ is holomorphic, it is bounded in the rectangle $|\text{Im}(z)| \leq 1$, and we need only control the behavior of that function for $|\text{Im}(z)| > 1$. If $\text{Im}(z) > 1$ and $z = x + iy$, then

$$\cot \pi z = i\, \frac{e^{i\pi z} + e^{-i\pi z}}{e^{i\pi z} - e^{-i\pi z}} = i\, \frac{e^{-2\pi y} + e^{-2\pi i x}}{e^{-2\pi y} - e^{-2\pi i x}},$$

and in absolute value this quantity is bounded. Also

$$\frac{1}{z}+\sum_{n=1}^{\infty}\frac{2z}{z^2-n^2}=\frac{1}{x+iy}+\sum_{n=1}^{\infty}\frac{2(x+iy)}{x^2-y^2-n^2+2ixy};$$

therefore if $y>1$, we have

$$\left|\frac{1}{z}+\sum_{n=1}^{\infty}\frac{2z}{z^2-n^2}\right|\leq C+C\sum_{n=1}^{\infty}\frac{y}{y^2+n^2}.$$

Now the sum on the right-hand side is majorized by

$$\int_0^{\infty}\frac{y}{y^2+x^2}\,dx\,,$$

because the function $y/(y^2+x^2)$ is decreasing in x; moreover, as the change of variables $x\mapsto yx$ shows, the integral is independent of y, and hence bounded. By a similar argument Δ is bounded in the strip where $\mathrm{Im}(z)<-1$, hence is bounded throughout the whole strip $|\mathrm{Re}(z)|\leq 1/2$. Therefore Δ is bounded in $\mathbb{C}$, and by Liouville's theorem, $\Delta(z)$ is constant. The observation that Δ is odd shows that this constant must be 0, and concludes the proof of formula (4).

To prove (3), we now let

$$G(z)=\frac{\sin\pi z}{\pi}\qquad\text{and}\qquad P(z)=z\prod_{n=1}^{\infty}\left(1-\frac{z^2}{n^2}\right).$$

Proposition 3.2 and the fact that $\sum 1/n^2<\infty$ guarantee that the product $P(z)$ converges, and that away from the integers we have

$$\frac{P'(z)}{P(z)}=\frac{1}{z}+\sum_{n=1}^{\infty}\frac{2z}{z^2-n^2}.$$

Since $G'(z)/G(z)=\pi\cot\pi z$, the cotangent formula (4) gives

$$\left(\frac{P(z)}{G(z)}\right)'=\frac{P(z)}{G(z)}\left[\frac{P'(z)}{P(z)}-\frac{G'(z)}{G(z)}\right]=0,$$

and so $P(z)=cG(z)$ for some constant c. Dividing this identity by z, and taking the limit as $z\to 0$, we find $c=1$.

Remark. Other proofs of (4) and (3) can be given by integrating analogous identities for $\pi^2/(\sin\pi z)^2$ derived in Exercise 12, Chapter 3, and Exercise 7, Chapter 4. Still other proofs using Fourier series can be found in the exercises of Chapters 3 and 5 of Book I.

4 Weierstrass infinite products

We now turn to Weierstrass's construction of an entire function with prescribed zeros.

Theorem 4.1 *Given any sequence* $\{a_n\}$ *of complex numbers with* $|a_n| \to \infty$ *as* $n \to \infty$*, there exists an entire function* f *that vanishes at all* $z = a_n$ *and nowhere else. Any other such entire function is of the form* $f(z)e^{g(z)}$*, where* g *is entire.*

Recall that if a holomorphic function f vanishes at $z = a$, then the multiplicity of the zero a is the integer m so that

$$f(z) = (z - a)^m g(z),$$

where g is holomorphic and nowhere vanishing in a neighborhood of a. Alternatively, m is the first non-zero power of $z - a$ in the power series expansion of f at a. Since, as before, we allow for repetitions in the sequence $\{a_n\}$, the theorem actually guarantees the existence of entire functions with prescribed zeros and with desired multiplicities.

To begin the proof, note first that if f_1 and f_2 are two entire functions that vanish at all $z = a_n$ and nowhere else, then f_1/f_2 has removable singularities at all the points a_n. Hence f_1/f_2 is entire and vanishes nowhere, so that there exists an entire function g with $f_1(z)/f_2(z) = e^{g(z)}$, as we showed in Section 6 of Chapter 3. Therefore $f_1(z) = f_2(z)e^{g(z)}$ and the last statement of the theorem is verified.

Hence we are left with the task of constructing a function that vanishes at all the points of the sequence $\{a_n\}$ and nowhere else. A naive guess, suggested by the product formula for $\sin \pi z$, is the product $\prod_n (1 - z/a_n)$. The problem is that this product converges only for suitable sequences $\{a_n\}$, so we correct this by inserting exponential factors. These factors will make the product converge without adding new zeros.

For each integer $k \geq 0$ we define **canonical factors** by

$$E_0(z) = 1 - z \quad \text{and} \quad E_k(z) = (1 - z)e^{z + z^2/2 + \cdots + z^k/k}, \quad \text{for } k \geq 1.$$

The integer k is called the **degree** of the canonical factor.

Lemma 4.2 *If* $|z| \leq 1/2$*, then* $|1 - E_k(z)| \leq c|z|^{k+1}$ *for some* $c > 0$.

Proof. If $|z| \leq 1/2$, then with the logarithm defined in terms of the power series, we have $1 - z = e^{\log(1-z)}$, and therefore

$$E_k(z) = e^{\log(1-z) + z + z^2/2 + \cdots + z^k/k} = e^w,$$